Leveraging Mono-Modal Data to Generate Multimodal Representations for Industrial Anomaly Detection

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Abstract

Generating multimodal representations from a single modality presents a promising direction for enhancing classification tasks in industrial anomaly detection, especially when access to multiple sensors is limited. In this paper, we propose a fully automated generative pipeline that synthesizes multiple modality representations from RGB images without human intervention or refinement. The generated modalities include thermal infrared and depth map images using modality-specific generative models, textual captions generated through image captioning, audio signals derived via image-to-audio transformation, and corresponding spectrograms. Each generated modality is evaluated independently through a binary classification task using top-performing pre-trained classifiers. Experimental results show that several generated modalities outperform the original RGB data, highlighting the potential of modality expansion for improving classification performance in mono-modal industrial settings. The study establishes a foundation for future multimodal fusion strategies and semantic consistency evaluation across generated modalities.

CCS Concepts

• **Computing methodologies** → **Scene anomaly detection**; **Classification and regression trees**; • **Information systems** → **Presentation of retrieval results**; • **Applied computing** → *Computer-aided design*.

Keywords

Industrial Anomaly Detection, Multimodal Representations, Generative Models, Thermal Infrared, Depth Maps, Image Captioning, Audio Signals, Spectrograms, Binary Classification

ACM Reference Format:
Charbel El Achkar, Abdallah Makhoul, and Joseph Azar. 2018. Leveraging Mono-Modal Data to Generate Multimodal Representations for Industrial Anomaly Detection. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation emai (Conference acronym 'XX)*. ACM, New York, NY, USA, 10 pages. https://doi.org/XXXXXXX.XXXXXXX

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Conference acronym 'XX, June 03–05, 2018, Woodstock, NY
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ACM ISBN 978-1-4503-XXXX-X/18/06
https://doi.org/XXXXXXX.XXXXXXX

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1 Introduction

In recent years, the field of anomaly detection has gained significant attention due to its critical role in industrial automation, defect detection, and predictive maintenance [11, 47]. Various datasets have been introduced to facilitate the development of anomaly detection models, focusing primarily on mono-modal image-based approaches. While these datasets have enabled advancements in detecting irregularities in industrial components, they remain limited in their ability to leverage multiple modalities for improved classification. Recent progress in multimodal learning [1, 17, 37] suggests that incorporating additional modalities, such as thermal, depth, and spectrogram-based audio data, can enhance anomaly detection by capturing complementary information beyond visual cues [8, 38, 57]. However, the lack of multimodal datasets specifically created for industrial anomaly detection presents a challenge in evaluating the benefits of such approaches.

This study addresses this gap by proposing a methodology for generating multimodal datasets from existing mono-modal image datasets, enabling a structured evaluation of different modalities in industrial anomaly detection. Unlike prior work that primarily focuses on mono-modal vision-based detection, our approach systematically generates and evaluates additional modalities for their effectiveness in anomaly detection—including thermal imaging, depth estimation, raw WAV audio signals, and audio-derived spectrograms. This automated generation process minimizes human-induced bias, ensuring a seamless transformation while allowing for experimentation in scenarios where human verification is absent, addressing a critical gap in existing research methodologies. By doing so, we aim to examine the impact of multimodal data representations on anomaly detection performance and compare how each modality operates when used independently. Each generated modality is evaluated independently alongside the original dataset to quantify its contribution to anomaly classification.

The remainder of this paper is structured as follows: Section 2 explores existing anomaly detection datasets and reviews recent studies on the use of each modality in anomaly detection tasks. Section 3 presents the dataset selection process and details the methodology for generating additional modalities. Section 4 outlines the evaluation strategy and experimental setup, while Section 5 compares the classification performance of different modalities. Finally, Section 6 concludes the study and provides insights into potential future work.

2 Related Work

Anomaly detection datasets are essential for training and evaluating models that identify irregularities in machinery, production

lines, and manufacturing processes in industries [7, 47]. These datasets capture a variety of scenarios, from defective components to complex machine faults, providing a robust foundation for both supervised and unsupervised anomaly detection tasks [11, 20]. They enable dual-modal systems to interpret and label anomalies by combining image data with annotations, advancing the development of automated inspection and predictive maintenance systems [72].

Below, we list several prominent datasets tailored for industrial anomaly detection. Notable among these is KolektorSDD [60] and its expanded variant KolektorSDD2 [9], both specifically designed for surface defect detection and evaluation within production lines where subtle surface anomalies need to be accurately classified. For automotive manufacturing, BTAD (Bosch Tube Anomaly Detection) [44] dataset offers data focused on detecting irregularities in machined metal components, facilitating model training for complex geometries commonly found in this sector. In the domain of electrical component manufacturing, MIAD (Magnetic Tile Anomaly Detection) [5] provides crucial data for defect detection on magnetic tiles, a precision-dependent task given the small scale and subtlety of common defects. The VisA (Visual Anomaly) [80] dataset offers high-resolution images with texture-based defects, supporting the development of systems capable of discerning subtle surface anomalies across materials. Meanwhile, VAD (Visual Anomaly Detection) [4] extends its scope beyond industrial boundaries, encouraging research into generalizable anomaly detection methods. The CDS2K [16] dataset addresses needs in electronics manufacturing while focusing on micro-defects on circuit boards, where even minor irregularities can impact functionality. Also, the InsPlad dataset (Industrial Surface Defects Dataset) [66] broadens applicability across materials and defect types, supporting adaptable inspection systems within multi-material manufacturing environments. A foundational dataset within this domain is MVTEC-AD (MVTEC Anomaly Detection) [6, 7], one of the most widely recognized datasets in anomaly detection research. This dataset includes high-quality images across various industrial object categories, each annotated with defective and defect-free samples, establishing MVTEC-AD as a benchmark for evaluating model performance in anomaly detection tasks. Its diversity and precision have made it a gold standard in the field, setting a reliable baseline for developing and comparing industrial anomaly detection models.

It is important to note that all these datasets are mono-modal, containing only visual data. This mono-modal approach provides essential image-based information. However, it limits the datasets' applicability in multimodal research scenarios where textual descriptions would enhance anomaly interpretation and labelling, in addition to other available modalities.

To address this limitation, recent advances in large language models (LLMs) and multimodal models have enabled systems not only to detect anomalies but also to generate descriptive textual feedback. These capabilities are especially critical in dual-modal datasets, where combining visual and textual data significantly enhances interpretability. As a result, anomaly detection and image annotation have become essential components of industrial automation, enabling systems to detect irregularities in machinery while providing detailed and descriptive annotations. However, the methods used for anomaly detection and caption generation differ significantly depending on the specialization of the models employed. In the path of using caption generation models to generate a textual representation of unlabelled industrial anomaly detection datasets, we list the following well-known and well-used models. They are grouped into two categories. The first category consists of the models capable of generating labels directly, and the second category consists of the anomaly detection models that require additional components to achieve labelling.

Regarding the first category, some multimodal models are inherently designed for captioning. They use visual and textual training to generate accurate descriptions from images, including those that define industrial anomalies. The Bootstrapping Language-Image Pre-training (BLIP) [31] generates labels directly from images and has shown strong performance on image labelling benchmarks. It is suitable for fine-tuning industrial datasets to identify and describe anomalies. Microsoft's Florence model [76] is a multimodal system available through Azure that can generate detailed captions in various scenarios. Although it is not open source, it can be accessed through cloud APIs and customized for specific tasks, including anomaly descriptions. In addition, OpenAI's CLIP (Contrastive Language-Image Pre-training) [49] generates embeddings from images that can be translated into captions when combined with a language model such as GPT-4 [2]. This setup allows for captioning capabilities tailored to industrial contexts through additional fine-tuning. Still among the multimodal models designed for captioning, VisualGPT [12] integrates vision and speech understanding, making it suitable for generating captions directly on industrial images. This model can be fine-tuned to describe detected anomalies in machine parts with remarkable accuracy, enhancing its utility in specialized industrial contexts. Similarly, Google's PaLM-E model [15] is a versatile, large-scale multimodal system that processes both text and image input to generate relevant captions even in complex visual scenarios. Although primarily a research model, its adaptability suggests significant potential for industrial captioning applications. The combination of DreamBooth [54] and ControlNet [77] offers controlled and context-aware caption generation, which is particularly effective when refined on specific industrial data sets. While requiring a more complex setup, this pairing provides nuanced descriptions of context-specific anomalies, allowing for highly customizable output.

Regarding the second category, models focused on visual anomaly detection can be enhanced by integrating with language models to generate descriptive labels. Meta's Segment Anything Model (SAM) [25], for example, excels at precise segmentation, isolating areas of potential anomalies. While SAM does not generate labels, it can be used alongside models such as CLIP [49] to effectively label segmented regions. DINOv2 [10], known for robust feature extraction, is beneficial for anomaly detection on machine images. When integrated with a language model such as GPT-4 [2], it can translate extracted features into meaningful labels, enriching the interpretability of anomaly detection results. For fast object detection, YOLOv8 [64] stands out as it identifies machine parts and flags anomalies with speed and accuracy. Although it lacks intrinsic captioning capabilities, coupling YOLOv8 with a language model results in the generation of descriptive output. MAE (Masked Autoencoders) [19] offers a different approach by reconstructing masked regions of the image, helping to identify irregular patterns that may signal anomalies. When combined with a speech decoder, the

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MAE can convey these findings in detailed captions. Additionally, DeepSVDD [53] and PatchCore [52] provide few-shot anomaly detection, invaluable for sparse industrial datasets. When augmented with a text-generating model, these systems provide descriptive insights into detected irregularities, extending their applicability to detailed anomaly reporting in industrial settings.

Thermal infrared (TIR) imaging plays a critical role in anomaly detection by capturing temperature variations across surfaces, enabling the identification of defects that might not be visible in standard RGB imaging [21, 35]. Industrial applications widely use TIR for detecting overheating components, insulation failures, and structural weaknesses [30, 32, 57]. Recent advancements, such as ThermalGAN [27], have further improved the quality and interpretability of thermal images by generating high-resolution thermal representations from low-quality inputs, enhancing anomaly detection performance. However, while thermal imaging provides invaluable insights, it often benefits from integration with additional modalities to improve detection accuracy and reduce false positives.

Depth sensing complements thermal imaging by offering three-dimensional spatial data that enhances defect detection in manufacturing and structural integrity assessments [59]. Depth maps allow for precise analysis of object geometry, making them useful for identifying anomalies such as misalignments, dents, and surface deformations [59, 69]. The MiDas model [8, 51] has been widely employed for monocular depth estimation, generating detailed depth representations that assist in structural anomaly detection. Combining depth data with thermal and RGB imaging has proven effective in various industrial applications, allowing for a more comprehensive understanding of material defects. Furthermore, techniques such as data augmentation have been employed to enhance image datasets, improving the robustness of anomaly detection models [33, 78].

Beyond visual and depth-based modalities, audio signals are a powerful tool for detecting anomalies in mechanical systems by capturing irregular acoustic patterns [45, 67]. Analyzing sound variations enables the early identification of faults such as bearing wear, motor misalignment, and structural instability. State-of-the-art methods such as spectral audio-based anomaly detection leverage time-frequency representations to improve fault classification [38]. However, raw audio signals are often challenging to interpret, necessitating transformation into visual formats such as spectrograms for more effective analysis.

Spectrograms bridge the gap between audio and vision-based anomaly detection by converting sound signals into a time-frequency representation that can be processed using image-based deep learning models [24, 68]. Libraries like Librosa [42] provide robust tools for processing raw audio signals, generating spectrograms, and extracting relevant acoustic features. Spectrogram-based methods have demonstrated strong performance when applied to industrial sound datasets, particularly when integrated with deep learning architectures such as CNNs [28] and Transformers [65] [79]. The ability to combine audio-derived spectrograms with thermal and depth data opens new avenues for multimodal anomaly detection, reinforcing the need for further exploration in this space.

3 Multimodal Generation Methodology

In this section, we introduce the MVTEC-AD [7] dataset and justify its selection over other industrial anomaly detection datasets. Although inherently unsupervised, we approach it as a supervised dataset through a detailed process. Additionally, we outline our methodology and preprocessing techniques for generating visual, caption-based, and signal-based modalities. This process extends a mono-modal dataset into a comprehensive multimodal dataset, enabling a broader evaluation of anomaly detection approaches.

3.1 The MVTEC-AD dataset

The MVTEC-AD [7] dataset, widely recognized for industrial anomaly detection, consists of 6,612 images categorized into 15 objects and 5 texture classes, with pixel-precise defect annotations. It is structured into 3,629 defect-free images for training, 1,725 images (both defect-free and defective) for testing, and 1,258 ground-truth images. The object categories include bottle, cable, capsule, hazelnut, metal nut, pill, screw, toothbrush, transistor, and zipper, while the texture categories cover carpet, grid, leather, tile, and wood.

Designed for unsupervised learning, MVTEC-AD evaluates models by training exclusively on defect-free images and testing against both normal and defective samples. However, its structured design and precise ground-truth annotations also make it suitable for supervised learning approaches. The dataset captures many real-world industrial defects, from minor scratches and dents to more complex structural irregularities, ensuring high model precision and robustness.

What sets MVTEC-AD apart is its diversity and real-world applicability. It includes small, intricate components such as screws and transistors as well as larger objects such as bottles and capsules, making it highly representative of industrial inspection tasks. Its comprehensive labelling and extensive adoption in the anomaly detection community establish it as a benchmark dataset for validating and comparing detection methodologies.

3.2 Pre-Processing

The MVTEC-AD dataset, developed for unsupervised anomaly detection, is adapted for supervised binary classification to support anomaly detection tasks. A systematic preprocessing pipeline is implemented to prepare the dataset for supervised learning models and enable multimodal generation.

Initially, the images are sorted into three distinct directories—*train*, *test*, and *ground truth*—according to their respective object classes, streamlining subsequent processing steps.

Image resolution across the dataset varies among six dimensions: 700×700, 800×800, 840×840, 900×900, 1000×1000, and 1024×1024 pixels. For consistency in binary classification, all images are resized to 700×700 pixels, the lowest resolution present. Zero-padding is applied to images with larger dimensions to preserve the aspect ratio and prevent distortion. This process involves:

- Creating a 700×700 pixel black canvas.
- Centering the resized image on the canvas.
- Filling any remaining space with black padding.

A structured naming convention is introduced to facilitate future processing of all images. The naming format is as follows: `ObjectName_Category_Status_Number.png` , where:

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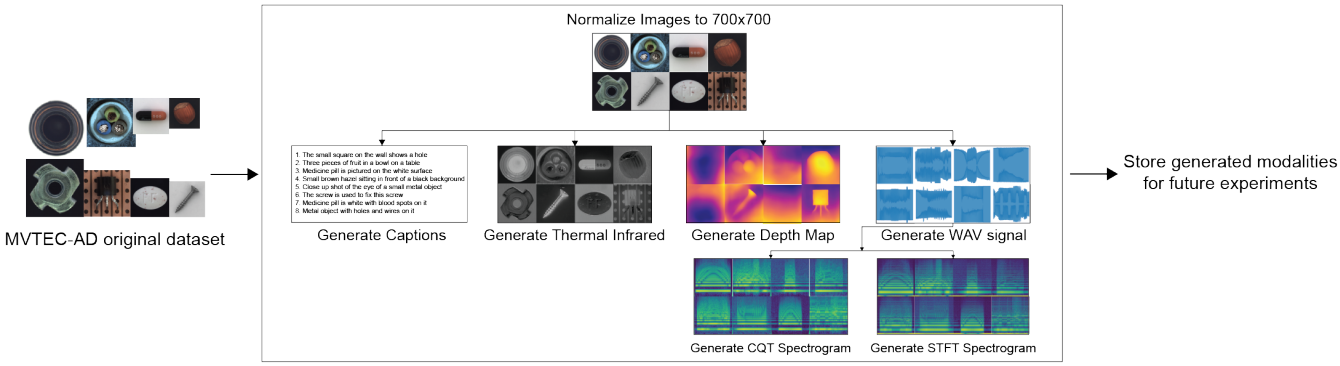

Figure 1: Pipeline for generating multiple modalities from a single image modality

- **ObjectName** specifies the object class (e.g., bottle, screw, transistor).
- **Category** indicates the dataset split (*train*, *test*, or *ground truth*).
- **Status** differentiates between defect-free and defective samples.
- **Number** corresponds to the original image identifier.

For example, an image initially named `000.png` is renamed to `bottle_train_good_000.png`, post-sorting to ensure clarity in dataset organization.
By the end of preprocessing, all images are resized to 700×700 pixels and systematically grouped into *train*, *test*, and *ground truth* directories. The latter serves as the foundation for modality generation and the development of robust supervised learning models for industrial anomaly detection.

3.3 Caption generation

Starting with our first modality, we intend to generate a caption for each image in the dataset, forming a text-based modality for the MVTEC-AD [7] dataset, and in this case, a caption representing what the image consists of. As shared in the related work section, there are many caption-generation models in the field. Some have built-in capabilities to generate captions directly, such as [31]. Others, like CLIP [49], are contrastive models designed to match images and text instead of creating captions. Therefore, these contrastive models can be used as feature-matching models and linked with GPT models [2, 50] to generate captions based on those extracted matching features. In this study, since we intend to experiment with multiple modalities, we chose to use BLIP [31] as a representative direct caption generation model and CLIP [49] as a contrastive model linked with GPT-2 [50] for caption generation.

Concerning BLIP [31], we tested the model on pre-processed images of size 700x700, where the captions were found to be highly inaccurate through human verification. Hence, the images were resized to 384x384, as BLIP's pre-trained models were designed and optimized for this specific size according to their documentation. Both the VIT-B_CapFilt-L and BLIP_VIT-L pre-trained models were used, and captions were generated for each image and stored in CSV for later use.

On the other hand, the CLIP [49] model was used as a contrastive model for feature extraction while being connected to a GPT-2 [50] model for caption generation. The CLIP model, available with Flickr30k [75], COCO [34], and Conceptual [56] datasets' pre-trained models, was used with all its variations alongside GPT-2, forming a wide range of generated captions. The combinations are listed below:

- **Feature Extractor (CLIP Pretrained Models):**
  - **Flickr30k:** RN50, RN101
  - **COCO:** ViT-B/16, ViT-B/32, ViT-L/14, ViT-L/14@336px, RN50, RN101, RN50x4, RN50x16, RN50x64
  - **Conceptual:** ViT-B/16, ViT-B/32, ViT-L/14, ViT-L/14@336px, RN50, RN101, RN50x4, RN50x16, RN50x64
- **Caption Generator:** GPT2-small, GPT2-large

Although the 700x700 images were appropriate for Flickr30k training, it was crucial to incorporate other dimensions, such as 224x224 for VIT-B/16/32 and VIT-L/14 training, and 336x336 for VIT-L/14@336. Other variations, such as 512x512, were also used whenever needed. Overall, we resized the images to match the size used by the pre-trained models to ensure the best outcome from the caption generation process. In summary, BLIP and CLIP's pre-trained models were employed to generate captions from the MVTEC-AD dataset. These captions are organized in CSV files, with each row containing the image name along with its corresponding caption, intended for subsequent use in the binary classification experiment.

3.4 Thermal Infrared generation

The caption modality helped us discover and generate a text-based dimension for the targeted dataset. However, additional correlated modalities can also be derived from an industrial image dataset, such as thermal infrared images, which capture temperature and heat variations from visual cues in the RGB images, simulating realistic thermal distributions. Studies such as TVA-GAN [74] and ParrotGAN [26], have explored generative adversarial networks for thermal image synthesis. TVA-GAN focuses on attention-guided cyclic GANs for thermal-to-visible transformations, emphasizing facial synthesis in low-light environments. ParrotGAN, on the other hand, leverages conditional adversarial networks for cross-modality person re-identification in aerial imagery. While these

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models demonstrate impressive capabilities in their respective domains, ThermalGAN [27] stands out for its efficiency and accuracy in translating RGB images into thermal infrared images, making it particularly suitable for industrial anomaly detection. ThermalGAN [27] is a generative adversarial network trained specifically to estimate thermal infrared properties from RGB input. It operates by learning the mapping between RGB and thermal domains, using paired datasets during its original training process. The generator in ThermalGAN predicts the thermal equivalent of an RGB image by leveraging learned features such as color, texture, and spatial information to approximate temperature distributions and heat patterns. Meanwhile, the discriminator ensures that the generated thermal images closely match the characteristics of real thermal infrared images. The grayscale intensity in these images serves as a proxy for thermal variations, where brighter regions suggest higher estimated temperatures and darker areas indicate lower ones, despite the absence of actual heat data (Figure 1). In an industrial setting, such generated thermal images can highlight structural and material differences, with uniform intensity implying consistent surfaces and deviations potentially indicating defects or inconsistencies.

In this study, the pre-trained ThermalGAN [27] model is used on the MVTEC-AD [7] dataset. Thermal infrared images were generated for each pre-processed 700x700 image in the dataset, enabling the introduction of a new thermal-based modality.

3.5 Depth Map generation

The generation of a depth map modality enables the extraction and generation of a spatial dimension for the targeted dataset. Beyond conventional 2D representations, depth maps provide insight into the three-dimensional structure of scenes by estimating the distance of objects from the camera. This estimation often relies on visual cues such as color gradients, texture variations, and shading patterns inherent in RGB images.

Various studies have explored monocular depth estimation techniques. For instance, methods like Shape-from-Shading analyze shading information to infer surface depth [41]. Similarly, Shape-from-Texture methods utilize texture distortions to gauge spatial configurations [40]. These approaches, however, can be sensitive to variations in lighting and surface properties, limiting their applicability across diverse scenarios.

In contrast, MiDaS [8], a state-of-the-art deep learning model, has demonstrated robust performance in monocular depth estimation across a wide range of scenes and conditions. Trained on a diverse set of datasets, MiDaS effectively captures depth information from single RGB images by leveraging learned features such as color, texture, and spatial relationships. The model's versatility and accuracy make it suitable for applications requiring detailed depth analysis. The generated depth maps are typically visualized using a false-color representation, where warmer colors (yellow and orange) indicate closer regions, while cooler colors (blue and purple) represent farther areas (Figure 1). This color encoding enhances spatial perception, allowing for a more intuitive interpretation of object distances within a scene.

In this study, we utilized the pre-trained MiDaS model to generate depth maps for the MVTEC-AD dataset. Each pre-processed 700x700 RGB image was processed to produce a corresponding depth map for the binary classification experiment.

3.6 Audio-Signal Generation

Transforming industrial images into audio signals introduces a novel modality for data representation, similar to reversing the spectrogram-to-wav process. This technique assigns a unique audio signature to each image in the MVTEC-AD dataset, providing an additional perspective for analysis. Such methodologies have been explored in assistive technologies, where visual information is converted into auditory cues to assist visually impaired individuals [58]. The Librosa [42] library, a comprehensive Python package for music and audio analysis, is implemented to achieve the image-to-audio conversion which involves several key steps:

(1) **Normalization:** Pixel values are scaled from [0, 255] to [-1.0, 1.0], aligning them with the amplitude range suitable for WAV audio samples.
(2) **Flattening:** Each image is transformed into a one-dimensional array, corresponding to a mono-channel audio waveform.
(3) **Scaling:** The data is adjusted to fit the 16-bit audio range, ensuring compatibility with standard audio formats.
(4) **Sample Rate Specification:** A sample rate of 44,100 Hz is applied, adhering to standard audio playback quality.
(5) **File Generation:** A WAV file is created by appending the '.wav' extension to the original image filename, resulting in a mono-channel audio file where each pixel's intensity is represented as an audio sample.

Utilizing Librosa's foundational functionalities enables reliable audio signal generation without artificial intelligence intervention, thereby mitigating the risk of false positives in modality creation. This method allows for the automated generation of the audio-based modality, streamlining the process of multimodality generation without human inspection and validation.

3.6.1 Signal to Spectrogram. Following the generation of audio signals, Short-Time Fourier Transform (STFT) and Constant-Q Transform (CQT) spectrograms are generated using Librosa [42]. These specific spectrograms are selected among many due to their proven efficacy in representing audio features pertinent to anomaly detection in industrial datasets. STFT provides a time-frequency representation by segmenting the signal into overlapping windows and applying the Fourier Transform to each segment. This method offers a balanced resolution in both time and frequency domains for capturing transient and stationary components within audio signals. Studies have demonstrated the utility of STFT-based spectrograms in detecting anomalies in industrial settings [18, 46]. Conversely, CQT offers a logarithmically spaced frequency analysis, providing higher resolution at lower frequencies and coarser resolution at higher frequencies. This characteristic aligns well with the harmonic and tonal structures often present in industrial audio signals, facilitating the detection of subtle anomalies that may be less discernible with linear frequency analyses. Research comparing STFT and CQT spectrograms indicates that CQT's variable resolution enhances the identification of low-frequency components, which are critical in industrial applications [3]. The STFT and CQT spectrograms, derived from WAV signals, leverage their complementary strengths to enhance anomaly detection and classification. These

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Table 1: Performance of Different Models on Various Image Modalities

Modality	Model Name	Orig. Size	Resized	Batch	Accuracy	Precision	Recall	F1-Score
Original Images	VIT-B-p16-384	700x700	384x384	16	0.76	0.48	0.05	0.10
	VIT-B-p16-384	700x700	384x384	8	0.72	0.26	0.08	0.12
	tf_efficientnet_b7_ns	700x700	600x600	8	0.90	0.74	0.87	0.80
	tf_efficientnet_b7	700x700	600x600	8	0.73	0.47	0.89	0.61
	ig_resnext101_32x48d	700x700	224x224	8	0.75	0.29	0.02	0.03
	tf_efficientnet_b7_ap	700x700	600x600	8	0.91	0.86	0.73	0.79
DepthMap Images	VIT-B-p16-384	700x700	384x384	16	0.24	0.24	1.00	0.39
	VIT-B-p16-384	700x700	384x384	8	0.76	0.00	0.00	0.00
	tf_efficientnet_b7_ns	700x700	600x600	8	0.69	0.38	0.41	0.40
	tf_efficientnet_b7	700x700	600x600	8	0.70	0.33	0.23	0.27
	ig_resnext101_32x48d	700x700	224x224	8	0.76	0.00	0.00	0.00
	tf_efficientnet_b7_ap	700x700	600x600	8	0.64	0.29	0.36	0.32
Thermal IR Images	VIT-B-p16-384	700x700	384x384	16	0.51	0.29	0.71	0.41
	VIT-B-p16-384	700x700	384x384	8	0.62	0.30	0.41	0.35
	tf_efficientnet_b7_ns	700x700	600x600	8	0.90	0.99	0.59	0.73
	tf_efficientnet_b7	700x700	600x600	8	0.86	0.69	0.77	0.73
	ig_resnext101_32x48d	700x700	224x224	8	0.73	0.34	0.11	0.16
	tf_efficientnet_b7_ap	700x700	600x600	8	0.88	0.91	0.54	0.68
CQT Spectrograms	VIT-B-p16-384	700x700	384x384	16	0.76	0.00	0.00	0.00
	VIT-B-p16-384	700x700	384x384	8	0.57	0.19	0.25	0.22
	tf_efficientnet_b7_ns	700x700	600x600	8	0.76	0.40	0.01	0.02
	tf_efficientnet_b7	700x700	600x600	8	0.74	0.45	0.44	0.45
	ig_resnext101_32x48d	700x700	224x224	8	0.75	0.42	0.06	0.11
	tf_efficientnet_b7_ap	700x700	600x600	8	0.80	0.64	0.36	0.47
STFT Spectrograms	VIT-B-p16-384	700x700	384x384	16	0.24	0.24	1.00	0.39
	VIT-B-p16-384	700x700	384x384	8	0.76	0.00	0.00	0.00
	tf_efficientnet_b7_ns	700x700	600x600	8	0.82	0.67	0.46	0.55
	tf_efficientnet_b7	700x700	600x600	8	0.81	0.70	0.35	0.47
	ig_resnext101_32x48d	700x700	224x224	8	0.76	0.00	0.00	0.00
	tf_efficientnet_b7_ap	700x700	600x600	8	0.82	0.71	0.44	0.55

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spectrograms will be evaluated in subsequent experiments to assess their impact as generated modalities within the binary classification experiment.

With the completion of the modality generation pipeline, the MVTEC-AD dataset has been successfully transformed into a multimodal framework. This pipeline covers the entire process, from preprocessing and normalization to generating caption-based, thermal infrared, depth map, and audio-based modalities. By incorporating STFT and CQT spectrograms, this generative approach enables extensive experimentation in multimodal data analysis and fusion.

4 Experimental Framework for Multimodal Evaluation

While the MVTEC-AD dataset is designed for unsupervised learning experiments, our pre-processing pipeline enables the generation of multiple new modalities while restructuring the dataset into a supervised learning-friendly format. The dataset is reorganised into three primary folders: train, test, and ground-truth. Initially, the training folder contained only non-defective images, while the test set included both defective and non-defective samples.

Since the objective is to perform supervised binary classification, the dataset is restructured and split after grouping and shuffling all available samples. The data is divided into 80% for training, 10% for testing, and 10% for validation, ensuring that defective samples are distributed proportionally across all subsets. This process ensures that the classifier is exposed to defective data during training, allowing it to learn meaningful patterns before being evaluated on unseen test samples. A CSV file that contains filenames (for images, audio, and captions), along with a binary label column (0 for non-defective, 1 for defective items) is generated to facilitate the training and testing operations. This standardized structure provides a uniform data format for all classifiers and allows seamless ingestion across different modalities. Additionally, each modality undergoes specific pre-processing steps such as image normalization, audio resampling, and text truncation to ensure consistency before training.

4.1 Model Selection and Implementation Details

For image-based anomaly classification, the top-performing classifiers are selected based on the timm library benchmark [70], prioritizing models with high accuracy. These models are trained on all image-based modalities, including the original RGB images, thermal infrared (TIR) images, depth map (DM) images, and spectrograms (CQT and STFT). The selected models include ViT-B-p16-384 [14], tf_efficientnet_b7_ns [73], tf_efficientnet_b7 [62], ig_resnext101_32x48d [39], and tf_efficientnet_b7_ap [71]. Selection criteria include benchmark performance, compatibility with pre-processed resolutions, and the potential to achieve high classification accuracy.

For text-based classification of generated captions, we use state-of-the-art large language models (LLMs), such as BERT [13], DistilBERT [55], RoBERTa [36], ALBERT [29], and GPT-4 [2]. Similarly, for audio classification, models that run directly on raw waveform data are selected such as RawNet [22] and RawNet2 [61]. Additional experiments were conducted with generative audio models like WaveNet [63], SampleRNN [43], and WaveRNN [23], but these models did not yield competitive results, leading the focus on RawNet-based classifiers for the audio modality.

All models are implemented using the PyTorch framework [48] and trained on a single Tesla V100 GPU with 32GB RAM. The batch size is chosen as the largest feasible size within the GPU's memory constraints, and the training process is controlled using an early stopping mechanism that halts training after five consecutive epochs without significant improvement. This ensures sufficient training while mitigating overfitting.

4.2 Evaluation Metrics for Industrial Anomaly Detection

The classification performance is evaluated using accuracy, precision, recall, and F1-score, as these metrics are well-suited for industrial anomaly detection tasks. Accuracy provides a general measure of classification correctness, while precision and recall ensure that defective samples are correctly identified without excessive false alarms. The F1-score is particularly relevant in industrial applications where defective samples are often underrepresented, making it essential to balance precision and recall. These metrics collectively provide a robust assessment of each model's effectiveness in identifying industrial defects.

5 Comparative Analysis of Modality Performance

This section presents a comparative analysis of model performances across different modalities, with the primary objective of assessing the best-performing models based on accuracy and F1-score. Each modality is evaluated independently to understand its behavior in isolation before exploring the potential of multi-modal fusion in future work. For the caption-based modality, only models with an accuracy of at least 75% and an F1-score above 15% were included, removing all ALBERT [29] and GPT-4 [2] results due to their 0% performance in at least one metric. The results highlight the differences in effectiveness among various architectures depending on the modality.

For image-based modalities (Table 1), EfficientNet-B7 AdvProp (tf_efficientnet_b7_ap) and EfficientNet-B7 Noisy Student (tf_efficientnet_b7_ns) consistently outperform other architectures across the Original, Thermal Infrared (IR), and Spectrogram-based images. In the Original Images category, EfficientNet-B7 AdvProp achieves the highest accuracy at 91% with an F1-score of 79%, followed closely by EfficientNet-B7 Noisy Student with 90% accuracy and an F1-score of 80%. DepthMap images present significant challenges, with most models failing to maintain high recall and F1-scores. EfficientNet-B7 Noisy Student achieves 69% accuracy with an F1-score of 40%, showing better generalization than other models, but overall, this modality remains difficult for classification. Thermal IR images show strong results for EfficientNet-B7 Noisy Student, which achieves 90% accuracy and an F1-score of 73%, followed by EfficientNet-B7 AdvProp at 88% accuracy and 68% F1-score. These findings suggest that these specific EfficientNet variants are particularly well-suited for infrared imagery analysis.

For spectrogram-based modalities, STFT representations perform slightly better than CQT spectrograms, likely due to differences in time-frequency resolution. The best-performing models for STFT spectrograms include EfficientNet-B7 Noisy Student and EfficientNet-B7 AdvProp, both achieving 82% accuracy with an F1-score of 55%. In the case of CQT spectrograms, EfficientNet-B7 AdvProp leads with an 80% accuracy and an F1-score of 47%, outperforming other models within this modality. Raw waveform classification using WAV signals (Table 2) lags behind, with RawNet2 (Batch 4) reaching 75% accuracy but a 0% F1-score, indicating a strong imbalance in predictions. RawNet (Batch 8) meets the threshold with 71% accuracy and an F1-score of 20%, highlighting the challenges of training deep architectures directly on waveforms without intermediate feature transformations like spectrograms.

For text-based modalities derived from image captions (Table 3), models utilizing CLIP-generated captions and GPT-2 embeddings perform competitively. RoBERTa (CLIP+GPT2-large) achieves the highest accuracy at 80% with an F1-score of 50%, followed closely by BERT (CLIP+GPT2-large) with 80% accuracy and an F1-score of 44%. DistilBERT (CLIP+GPT2-large) also performs well with 77% accuracy and an F1-score of 40%. These results demonstrate the effectiveness of textual representations in classification tasks, especially when leveraging strong pre-trained models for caption generation and interpretation.

Overall, EfficientNet-B7 AdvProp and EfficientNet-B7 Noisy Student demonstrate the most robust architectures for image-based and spectrogram-based tasks, while RoBERTa proves highly effective for text-based classification. Thermal IR images exhibit strong classification potential with EfficientNet models, whereas DepthMap images and raw waveforms remain challenging. The need for further improvements in feature engineering and multi-modal fusion techniques is evident, particularly for these more complex modalities. These findings highlight the need for adaptive architectures that can better handle the varying nature of these data representations, which is an avenue for future exploration.

6 Conclusion and Future Work

In this paper, a generative pipeline was proposed for learning multimodal representations from a single image-based dataset in the

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Table 2: Performance of Models on Audio Signal Modality

Modality	Model Name	Sample Rate	Audio Length	Batch	Accuracy	Precision	Recall	F1-Score
WAV Signals	RawNet	44100	11.11s	8	0.71	0.32	0.15	0.20
	RawNet	44100	11.11s	16	0.69	0.37	0.29	0.33
	RawNet	44100	11.11s	32	0.69	0.36	0.28	0.31
	RawNet2	44100	11.11s	4	0.75	0.00	0.00	0.00

Table 3: Performance metrics for different models and captioning methods

Modality	Model Name	Generated Captions	Batch Size	Accuracy	Precision	Recall	F1-Score
Captions	BERT	BLIP_VIT-B_CapFilt-L	128	0.75	0.48	0.21	0.29
		CLIP(Flickr30k)+GPT2-large	128	0.80	0.75	0.31	0.44
		CLIP(Flickr30k)+GPT2-small	128	0.77	0.67	0.21	0.32
		CLIP(RN50x16+COCO)+GPT2	128	0.76	0.56	0.22	0.31
		CLIP(RN50x64+Conceptual)+GPT2	128	0.76	0.60	0.16	0.25
		CLIP(ViT-B-16+COCO)+GPT2	128	0.78	0.66	0.26	0.38
		CLIP(ViT-B-16+Conceptual)+GPT2	128	0.77	0.64	0.21	0.31
		CLIP(ViT-L-14@336px+COCO)+GPT2	128	0.79	0.75	0.23	0.35
		CLIP(ViT-L-14@336px+Conceptual)+GPT2	128	0.78	0.61	0.33	0.43
		CLIP(ViT-L-14+COCO)+GPT2	128	0.77	0.55	0.34	0.42
		CLIP(ViT-L-14+Conceptual)+GPT2	128	0.79	0.76	0.22	0.34
Captions	DistilBERT	CLIP(Flickr30k)+GPT2-large	128	0.77	0.59	0.30	0.40
		CLIP(RN50x16+COCO)+GPT2	128	0.77	0.65	0.16	0.26
		CLIP(ViT-B-16+COCO)+GPT2	128	0.79	0.76	0.23	0.35
		CLIP(ViT-B-16+Conceptual)+GPT2	128	0.77	0.59	0.32	0.42
		CLIP(ViT-L-14@336px+COCO)+GPT2	128	0.78	0.59	0.45	0.51
		CLIP(ViT-L-14@336px+Conceptual)+GPT2	128	0.76	0.52	0.42	0.47
		CLIP(ViT-L-14+COCO)+GPT2	128	0.77	0.55	0.42	0.47
		CLIP(ViT-L-14+Conceptual)+GPT2	128	0.79	0.71	0.30	0.42
Captions	RoBERTa	CLIP(Flickr30k)+GPT2-large	64	0.80	0.69	0.39	0.50

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context of industrial anomaly detection. The pipeline relies on modality-specific generators to create thermal infrared and depth map images from the original RGB modality. Caption models were then employed to generate textual descriptions as a text-based modality. Additionally, an audio-based modality was derived using an image-to-audio transformation, and its spectrograms were extracted as additional representations. Each generated modality was evaluated independently through a binary classification task using top-performing pre-trained classifiers. The entire process of modality generation and classification was performed without any human intervention or manual refinement. The results showed that some generated modalities achieved higher classification performance than the original modality, highlighting the potential of generative modality expansion for improving binary classification tasks in mono-modal industrial settings. Our future work will explore late fusion strategies, such as retrieval-augmented generation (RAG), to combine the decision boundaries of multiple generated modalities. Furthermore, it will be essential to evaluate the semantic alignment and consistency between the generated modalities and the original data using cross-modal similarity or retrieval-based techniques. We also aim to integrate the pipeline into a fully end-to-end trainable system and expand the evaluation to more complex industrial datasets and real-world anomaly detection scenarios.

Acknowledgments

Computations have been performed on the supercomputer facilities of the "Mésocentre de Franche-Comté".

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This paper is a work in progress. Not for distribution.